Methodology — the Yvynation batch pipeline and analytical framework

This note delineates *how* the four-territory results were produced, so the chapter narratives can be read against a reproducible method. Everything here is driven by the batch module `yvynation/pages/batch_processing.py` and its backend handlers in `state/`, running against Google Earth Engine (EE).

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1. What the framework is trying to measure

The core empirical claim is not simply *"forest cover is higher inside an Indigenous Territory (TI) than outside"* — that could be self-selection, since remote lands were often demarcated *because* they were remote. The claim is that the **rate of land-cover change inside the boundary differs systematically from the rate in an adjacent, unprotected strip** that shares the same biome, road network, commodity market, and climate. The 10 km exclusive buffer is constructed to be that counterfactual (see §5).

A second claim is about **policy lag**: the sharpest inflection in a land-cover trend does not coincide with the year of the policy event that (hypothetically) caused it. The framework treats that lag as finite, measurable, and causally informative — and, as it turns out, of three distinct kinds (see SYNTHESIS.md §3).

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2. Data sources

Source	Product	Role
MapBiomas	Collection 9 (1985–2023) for the April run; Collection 10 (1985–2024) for the May run	Annual land-cover *class* areas and class-to-class transitions
Hansen GFC	Global Forest Change (v1.11 for April; v1.12 / `2024_v1_12` for May)	Tree-cover 2000 baseline, annual canopy **loss** (2001–), gain (2000–2012)
Hansen / GLAD	GLCLU 2020	A second, independent per-class forest-cover snapshot
FUNAI	Indigenous Lands boundaries (657 territories)	Territory geometry source
CNUC	Conservation Units (3,247 units)	Alternative geometry source (same pipeline)

Using **two independent forest measures** (MapBiomas class change vs Hansen canopy loss) is deliberate: the gap between them is part of the validation story, not noise. The clearest example is Kayapó, where MapBiomas reports −1.9% Forest Formation over 1985–2023 but Hansen reports a 14.2% canopy-loss footprint (§9.12.2) — the divergence flags fire/degradation that MapBiomas' persistence logic re-absorbs into the same class.

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3. The pipeline (what a batch run does)

A run is configured entirely in the left/right panels of `batch_processing.py` and then executed unattended by `AppState.run_batch_processing`. For each selected territory (and, optionally, its buffer) the following analyses can be switched on:

Analysis toggle	Output
🌿 MapBiomas single-year	Class composition + distribution for one year
📊 Year-over-year comparison	Δha / Δ% / gains–losses between `year` and `year2`
🟦 Class-transition treemaps	Faceted treemap per class (small classes → "Others")
🌲 Hansen GLAD forest cover	GLCLU 2020 per-class breakdown
🪓 Hansen GFC (loss / gain)	Tree-cover 2000, annual loss series, gain
🗺️ PNG maps & charts	Satellite + MapBiomas y1/y2 raster PNGs
🌀 Multiple time-window	Sankey + Sunburst + treemaps across N stages
📈 Deforestation timeline	Hansen + MapBiomas + fire, with policy/political context bands

Extra MapBiomas auxiliary rasters (rendered for `year2`) can be added on top of the PNG maps: **deforestation & secondary vegetation**, **annual burned area**, **fire frequency (full 1985–2024 period)**, **year of last fire**, **mining substances**, and **agriculture cycles**. These are what produced the fire-scar and mining overlays discussed in the Kayapó chapter (§9.6, §9.12).

The multi-time-window mode mimics the original offline Python code: either a **constant step** (1, 2, 4, 5 or 8 years, with 1985 → 2024 forced as endpoints) or a **custom** list of 3–4 comma-separated years (e.g. `1985, 2004, 2012, 2023` to align stages with policy events). Each stage gets its own Sankey/Sunburst so a single figure can carry the whole 40-year trajectory.

Expected cost is **2–10 minutes per territory** depending on toggles; the tab runs in the background and can be stopped after the current territory.

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4. Output artefacts (what a run produces)

Everything is packaged into one self-describing ZIP. Per territory the folder contains:

```
metadata.json          territory id, source, year pair, run timestamp,
analysis_type
README.txt             human-readable analysis summary
figures/               11 interactive HTML figures (Plotly)
territory/<name>/       per-class CSVs + boundary.geojson +
transitions.json
```

The **11 figures** (identical filenames across territories, so cross-case comparison is trivial):

```
mapbiomas_composition.html      class composition, latest year
mapbiomas_distribution.html     class-area bar chart, latest year
change_percentage.html          class-by-class % change y1→y2
gains_losses.html               diverging bar of net gains/losses
transitions_matrix.html         source-class × target-class heatmap
transitions_sankey.html         flow diagram of the same transitions   ★
headline figure
transitions_sunburst.html       hierarchical transition view
hansen_glad_distribution.html   GLAD GLCLU 2020 per-class breakdown
hansen_gfc_summary.html         Tree Cover 2000 + Loss + Gain panel
hansen_gfc_loss_by_year.html    annual canopy-loss series             ★
policy-lag figure
hansen_balance.html             net forest balance
```

The **CSVs** are the ground truth the chapters quote. Example, the Kayapó `comparison_1985_vs_2023.csv` carries one row per class:

```
Class_ID, Area_Year1, Class_Name, Area_Year2, Change_ha, Change_km2,
Change_pct, Abs_Change
3,  2,875,570, Forest Formation, 2,821,132, −54,438, ..., −1.89%, ...
15,     2,105,   Pasture,              30,597, +28,492, ..., +1352.58%, ...
30,       951,   Mining,               13,825, +12,874, ..., +1352.56%, ...
```

`hansen_gfc_summary.csv` gives the three headline Hansen numbers (Tree Cover 2000, Forest Loss %, Forest Gain %); `hansen_gfc_loss_by_year.csv` gives the annual series; `transitions.json` gives the normalised class→class flow weights behind the Sankey.

A post-processing step (`_build_bundle.py`, `_export_pngs.py` in the results folder) renders all HTML figures to a paginated PDF and to 2800×1800 PNGs, both flat and per-territory, for dropping into the thesis document.

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5. The buffer as counterfactual

When the buffer toggle is on, the same analysis runs on an **external ring** (default 10 km) around each territory, written to `buffer/{territory}_Buffer_{km}km/`. The ring is *not* clipped to remove other protected areas inside it, so the comparison is deliberately territory-vs-whatever-is-actually-there (unprotected land *and* adjacent protected land).

The logic: where the same highway runs along the boundary, interior and buffer pixels face the same accessibility; where the same soy frontier expands, both face the same commodity price; where the same drought reduces fire resistance, both face the same climate shock. The **only systematic difference is legal status** and the institutional response it triggers. Buffer results are summarised in `multi_window.md` and discussed in SYNTHESIS.md §4.

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6. Quadrant clipping for very large areas

Kayapó (~3.3 M ha reported extent, part of an 11 M ha contiguous block) cannot be processed as one EE call without timing out. The batch clips any geometry larger than **1 million hectares into four quadrants** (NW, NE, SE, SW), processed and stored separately. This was a computational necessity that turned into an analytical asset: the quadrants isolate pressure fronts — the **eastern** quadrants border rural properties (pasture, agriculture, mining), while the **western** quadrants adjoin other protected areas and harder-to-reach zones. The NE quadrant is the mining hot-spot; the SW is nearly stationary (see `multi_window.md` and RESULTS §Kayapó).

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7. The two data epochs (why numbers differ)

Run	MapBiomas	Hansen	End year	Where reported
April 2026 (per-territory folders)	Collection 9	GFC to 2024	2023	`RESULTS.md`, chapters §7.10–§10.10
May 2026 (batch + buffer + multi-window)	Collection 10	GFC to 2025	2024	`multi_window.md`, report batch section

The report is explicit (§11.1) that Table 11.1 and the chapter figures use slightly different inputs. The gap is *informative*: it isolates which findings depend on the choice of headline metric (MapBiomas class change vs Hansen canopy loss) and which survive both. Collection 11 (1985–2025, expected August 2026) will trigger a reconciliation pass.

The Hansen-2024 caveat

The April Kayapó export reported **305,250 ha of 2024 loss** — roughly two-thirds of the entire 2001–2023 cumulative loss in a single year (see `hansen_gfc_loss_by_year.csv`). This is implausible as canopy *removal* and reflects a **2024 wildfire scar** captured by the `2024_v1_12` asset; MapBiomas Collection 10 independently reports >300,000 ha of post-wildfire canopy loss in the same window. The figure is reported as-is but flagged, and it is the main reason the project moved to Collection 10 and will move to Collection 11.

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8. Reproducing any of this

Every chart and table in the report can be regenerated from the app for **any** FUNAI territory or (with minor changes) CNUC conservation unit:

1. Open the batch page, pick source type, search and tick territories.
2. Set MapBiomas `year`/`year2` and the Hansen GLAD reference year.
3. Enable the analyses you want (+ auxiliary rasters, + multi-window, + timeline).
4. Optionally enable the 10 km buffer.
5. Start, wait, download the ZIP.

The individual-territory module offers the same analysis with an interactive map; the batch module is the reproducible, headless path for a known set of areas.